PHD RESEARCH HANDBOOK · UNIVERSITI SAINS MALAYSIA

Consumer AI Scoring of IGCSE Summary Writing

Consistency, Agreement and Construct Validity for Multilingual Learners

A complete, step-by-step operating manual for conducting the study — from preparing the instruments to submitting the thesis — written so that someone new to the project could carry it out.

RESEARCHER
Ehab Hourani

PROGRAMME
PhD (Language Studies), by research

SCHOOL
Languages, Literacies and Translation

CANDIDATURE
Sep 2027 – Aug 2031

Set-up | Data collection | Analysis | Writing

How to use this handbook

This handbook turns the research proposal into actions. The proposal explains *why* the study is designed as it is; this handbook explains *exactly what to do, in what order, with what materials*. Each stage has the same structure: a coloured header showing when and who, the purpose, numbered steps, the rules that must not be broken, and an end-of-stage checklist.

Golden rule

Anything marked ◆ **Preregistered rule** is fixed before main data collection. Do not change it afterwards without recording a dated deviation note and telling your supervisor.

If you are new to the project

Read Section 1 (the study on one page), then Section 2 (key terms), then go to the stage you are working on (Stages 0 and A–I). The Supervisor Q&A (Section 5) is the fastest way to understand the reasoning behind each choice.

Contents

1 The study on one page

2 Key terms in plain language

3 Master timeline & workload

4 People and roles

0 Stage 0 — Preparation before enrolment

A Stage A — Instruments (Sep–Dec 2027)

B Stage B — Ethics & permissions

C Stage C — Pilot & preregistration (Jan 2028)

D Stage D — Student writing & corpus (Feb–Mar 2028)

E Stage E — Human marking (Apr–May 2028)

F Stage F — AI scoring (winter break 2028–29)

G Stage G — Quantitative analysis (RQ1, RQ2)

H Stage H — Rationale analysis (RQ3)

I Stage I — Synthesis, papers & thesis

5 Supervisor questions & answers

6 All decision rules in one place

7 Risk register

8 Templates & forms

9 Data management & file naming

1 The study on one page

Teachers paste student writing into ChatGPT or Claude and ask for a mark. This study tests whether those marks are **repeatable**, whether they **agree with experienced human markers**, and whether the **reasons the AI gives** respect the difference between reading and writing in the IGCSE summary task.

65–80
students (4 IGCSE classes, Erbil)

130–160
scripts (2 passages each)

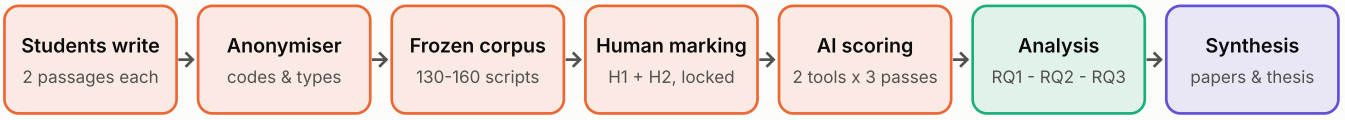
2 × 3
AI tools × scoring passes

780–960
main AI scores (+60 anchors)

2
human raters, blind & independent

8
scripts / 16 AI rationales coded (RQ3)

HOW DATA MOVES THROUGH THE STUDY



RESEARCH QUESTIONS

RQ	Question (short form)	Main evidence
RQ1	Do ChatGPT and Claude give the same script the same mark across three independent passes? Does this differ by passage?	ICC(1,1) across passes; per-script SD
RQ2	How closely do AI marks agree with two experienced human raters, and where (criterion, score level, passage) do they diverge?	QWK, ICC(A,1), correction gap, Bland–Altman
RQ3	In selected high-divergence and close-agreement cases, are the AI's reasons supported by the text, and do they respect the reading/writing distinction?	Claim-level evidence checking; dual coding
Synthesis	What does the combined evidence say about using AI as a second marker, and what still needs validating?	Summary across RQ1–RQ3

THE TASK BEING MARKED

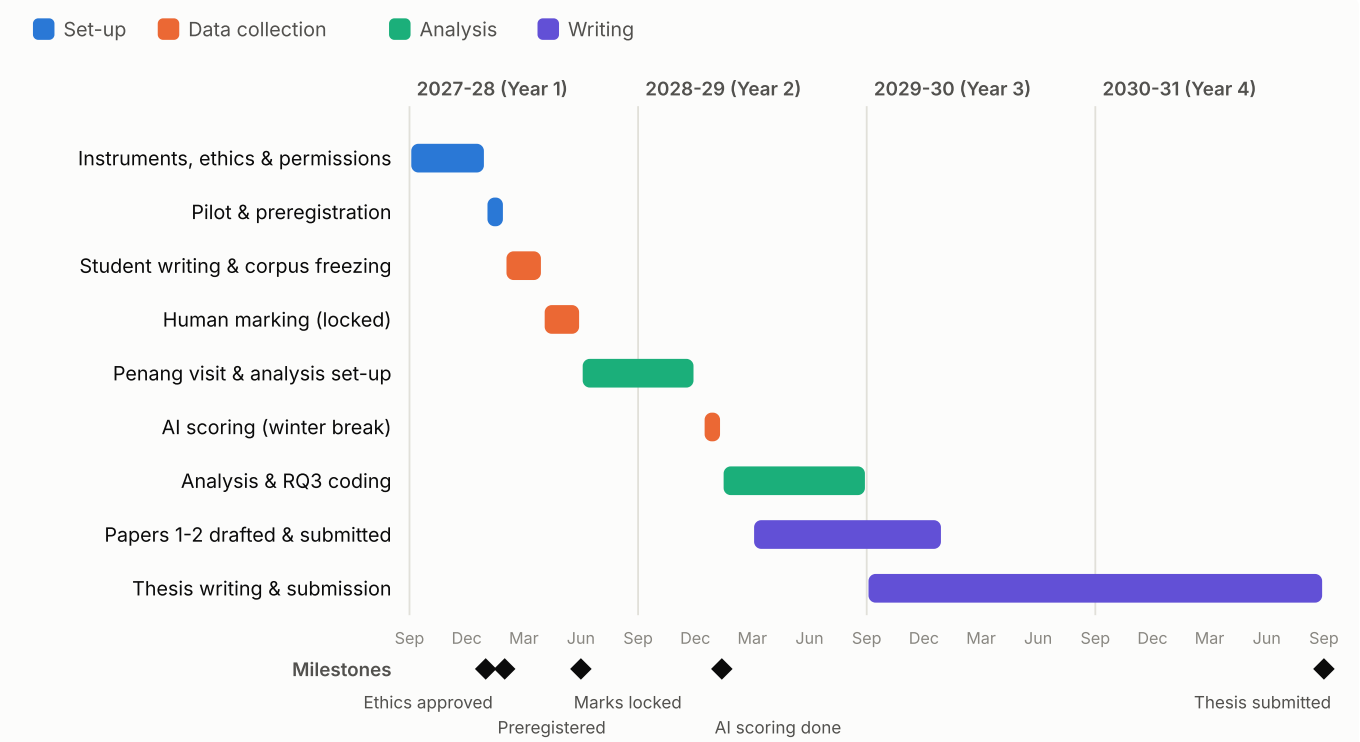
Cambridge IGCSE First Language English 0500, Paper 1, **Question 2(a)**: a selective summary of a passage, in the student's own words, continuous writing, **no more than 120 words**. Marked **out of 10 for Reading** (Table A: five levels of two marks, plus 0) and **out of 5 for Writing** (Table B: three levels — 4–5, 2–3, 1 — plus 0).

2 Key terms in plain language

Term	Meaning
Script	One student's summary of one passage. Each student produces two scripts (Passage A and Passage B).
Pass	One complete round in which every script is scored once by a tool. There are three passes per tool.
Rubric-embedded framing	The single fixed AI prompt: passage + task + student summary + study rubric + reading key + required output line (Appendix A of the proposal; Template T1 here).
Pass-to-pass consistency	How similar a tool's marks for the same script are across the three passes (RQ1).
H1, H2	Human Rater 1 (you, the candidate) and Human Rater 2 (an experienced IGCSE colleague).
Correction gap	AI mark minus the mean of H1 and H2. Positive = AI more lenient; negative = AI harsher.
QWK	Quadratic weighted kappa: agreement between two sets of ordered marks, giving partial credit for near-misses.
ICC	Intraclass correlation coefficient. ICC(1,1) = repeatability across passes; ICC(A,1) = absolute agreement between AI and one human rater.
Anchor scripts	Three pilot scripts (low, mid, high) scored by both tools at the start of every scoring day to monitor day-to-day stability.
Construct confusion	The AI uses a quality belonging to one criterion to justify the other criterion's mark — e.g., giving reading credit because the writing is fluent.
Source-fidelity error	The AI misdescribes the content — e.g., credits a point the student did not include.
Non-standard but meaningful phrasing	Wording that departs from conventional English but still conveys the source meaning accurately. Not attributed to any home language.
Preregistration	A time-stamped public record (e.g., on OSF) of the analysis plan, made before main data collection.
Corpus freezing	The point after which typed scripts can no longer be changed and the identity linkage key is destroyed.
Claim	One separate statement inside an AI rationale, e.g., "the student mentions rising costs". The unit of RQ3 coding.

3 Master timeline & workload

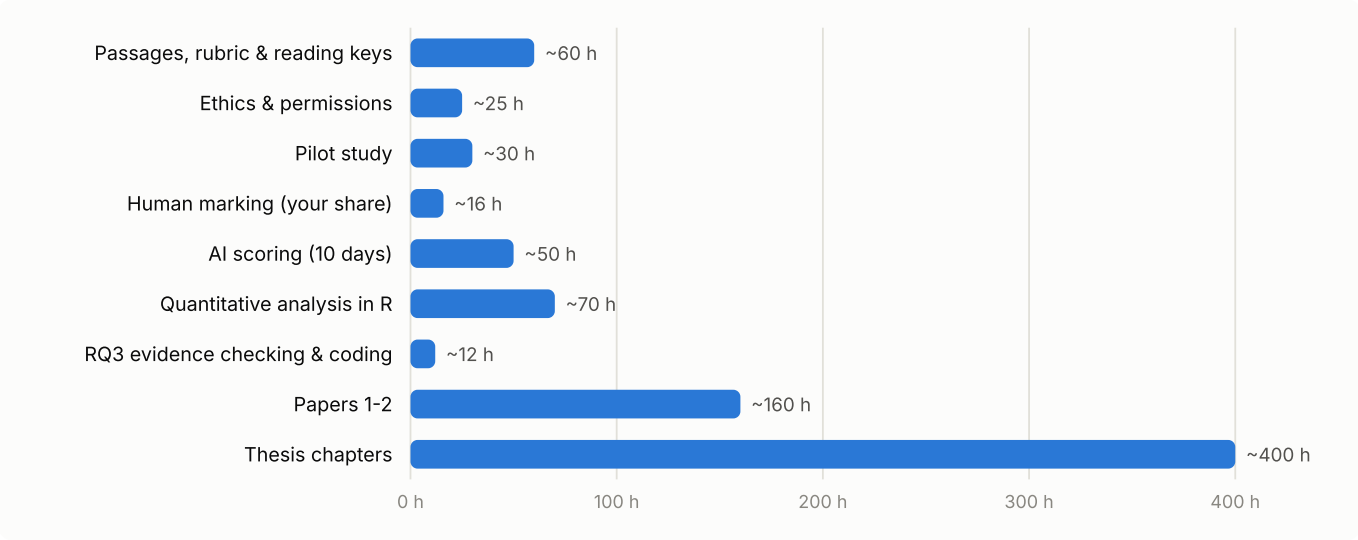
The candidature runs from September 2027 to August 2031. All student data are collected in **Year 1**; AI scoring is done in the **winter break of Year 2**, after human marks are locked.



When	What happens	Output / milestone
Sep–Dec 2027	Submit ethics application in the first weeks; write Passages A and B, rubric, reading keys; authenticity check; criterion-mapping review; KRG/school permissions	Ethics approval; final instruments
Jan 2028	Pilot with colleague's class (third passage); check timings, prompt, H1–H2 agreement; choose anchors; preregister	OSF preregistration
Feb–Mar 2028	Students write both summaries (counterbalanced order); anonymiser codes, types, re-checks	Frozen corpus; linkage key destroyed
Apr–May 2028	Calibration; blind independent marking by H1 and H2 (from ≥ 4 weeks after last writing session)	Locked human marks
Jun–Nov 2028	Summer supervision visit (Penang); prepare and test R scripts on pilot data	Tested analysis code
Dec 2028–Jan 2029	AI scoring: 3 passes × 2 tools in 10 working days (winter break)	Complete session log
Jan–Aug 2029	RQ1–RQ2 analysis; RQ3 selection, evidence checking, dual coding	Results; Papers 1–2 drafted
Sep 2029–Aug 2031	Thesis writing; optional Paper 3	Thesis submitted

Estimated hours for you, by stage

Planning estimates only — adjust after the pilot. Your colleague (Rater 2) needs roughly 13–16 h of marking plus 8–10 h of RQ3 coding.



Weekly load against your 15-hour allowance

Period	Typical hours / week	Main work
Sep–Dec 2027	8–12	Ethics submission, passages, rubric, reviews
Jan 2028	10–15	Pilot, timings, preregistration
Feb–Mar 2028	3–5	Supervising writing sessions (anonymiser types)
Apr–May 2028	4–6	Your share of blind marking
Jun–Nov 2028	5–10	Penang visit; R scripts tested on pilot data
Winter break 2028–29	**~25 (two weeks only)**	AI scoring — deliberately placed in a holiday
Jan–Aug 2029	10–15	Analysis, RQ3 coding, Papers 1–2
Sep 2029–Aug 2031	10–15	Thesis

The one intensive fortnight

AI scoring is the only period that exceeds your weekly allowance. Protect those ten working days completely: no marking, no other projects, and tell family and school in advance.

4 People and roles

Keep the team small. Each person below has a clearly bounded job; nobody except you needs to work on the project for more than a few days in total.

Role	Who	What they do	Time
Candidate / H1	Ehab Hourani	Writes instruments; runs pilot; marks all scripts as Rater 1; runs all AI scoring; analysis; RQ3 coding; writing	Throughout
Rater 2 (H2)	Experienced IGCSE colleague	Calibration; blind marking of all scripts; independent RQ3 evidence checking and coding on the same claim list; one disagreement meeting	~25–30 h
Anonymiser	Trusted non-teaching person (e.g., admin staff)	Holds consent records and originals; assigns codes; removes identifying content; types scripts verbatim; re-checks; destroys linkage key	~15–20 h
Pilot colleague	IGCSE teacher of another class	Lets 10–15 of their students write the pilot summary	1 lesson
Authenticity reviewers	2–3 IGCSE colleagues	Rate study passages vs real past passages; two of them review the criterion-mapping table	~2 h each
Supervisor	USM main supervisor	Approves instruments and preregistration; reviews final RQ3 coding of 3 scripts; decides on deviations	Meetings
Statistical adviser	USM statistician (via supervisor)	Reviews precision estimate and mixed-model specification before preregistration	1–2 meetings
Principal	School principal	Gives written permission; supports scheduling of writing sessions	Brief

Separation of roles matters

You are the students' teacher, the researcher and Rater 1. The anonymiser, not you, must collect consent, hold originals and type scripts. You must never see handwriting or names. This is what makes your marking blind.

0 Preparation before enrolment

When: Now → Aug 2027 **Who:** You

Nothing here collects data. These steps make Year 1 run on time.

1. **Secure a supervisor.** Send a 2-page synopsis + CV + full proposal to relevant USM staff (Applied Linguistics; Testing & Evaluation). Complete the USM application.
2. **Recruit the team informally.** Ask Rater 2, the anonymiser and the pilot colleague now; explain the time each role needs (Section 4).
3. **Talk to the principal.** Obtain informal agreement; ask whether KRG Ministry of Education or school-owner permission is required.
4. **Draft ethics documents** (Template list in Stage B) so they can be submitted in the first weeks of candidature.
5. **Start writing the passages** (Stage A specification). Early drafts reduce pressure in Sep–Dec 2027.
6. **Build skills:** install R + RStudio and work through a basic tutorial; install the free Atlas.ti student version; read the key methods sources (Kane 2013; Koo & Li 2016; Williamson et al. 2012; Sawaki et al. 2025).

- [ ] Supervisor agreed and application submitted
- [ ] Rater 2, anonymiser and pilot colleague have agreed in principle
- [ ] Principal agrees in principle; permission route known
- [ ] Draft consent/assent/information sheets exist
- [ ] First drafts of Passages A and B exist
- [ ] R, RStudio and Atlas.ti installed

A Instruments: passages, rubric, reading keys

When: Sep – Dec 2027 **Who:** You + colleagues

A1. Write the two source passages

Specification	Requirement
Length	300–400 words each
Text type	Non-fiction, modelled on 0500 Paper 1 Text B (e.g., article, report, opinion piece)
Content points	A clear, countable set of relevant points (the specimen has 12) plus some irrelevant detail, so that *selection* is tested
Difference between A and B	Different topic and different text structure, so passage effects are interpretable
Originality	Unpublished, written by you; never posted online before scoring is complete
Task wording	"According to the text, what are ...? Use continuous writing (not note form) and use your own words as far as possible. Your summary should not be more than 120 words."

A2. Write the reading key for each passage

1. List every acceptable content point in the order it appears, with acceptable paraphrases.
2. Mark which details are irrelevant to the task focus.
3. Add the instruction that other valid ideas may be credited (as in the Cambridge mark scheme).
4. Ask Rater 2 to check each key independently; resolve differences before the pilot.

A3. Write the study rubric (no verbatim Cambridge text)

Component	Structure (parallel to Cambridge)	What the levels describe
Reading /10	Five levels: 9–10, 7–8, 5–6, 3–4, 1–2; plus 0	Range of relevant ideas understood; focus on text and task; selection (from indiscriminate / excess material to skilful selection showing an overview)
Writing /5	Three levels: 4–5, 2–3, 1; plus 0	Clarity and fluency of expression; effectiveness of organisation; concise use of a range of appropriate vocabulary; reliance on the words of the text / lifted sections at lower levels

Never move a quality to the wrong criterion

Do not add 'grouping or linking points' to Reading (organisation belongs to Writing). Do not treat borrowed wording as poor selection (it belongs to Writing). Do not penalise necessary terms or names that have no reasonable alternative.

A4. Structured passage-authenticity check

1. Choose 2–3 real published 0500 passages from pre-2027 papers. **These are never submitted to any AI tool.**
2. Give 2–3 IGCSE colleagues all passages (study + real), unlabelled, with Template T5.
3. They rate each on difficulty, text type, content-point density, vocabulary level and length (1–5).
4. Compare ratings: study passages should fall within the range of the real passages. Revise if not.

A5. Criterion-mapping review

Two of the same colleagues judge each row of the mapping table (proposal Appendix C; Template T6) as **full / partial / absent**. Any row judged partial or absent by either reviewer is revised and re-reviewed before the pilot.

- [ ] Passages A and B final (300–400 words, countable points + irrelevant detail)
- [ ] Reading keys checked by Rater 2
- [ ] Study rubric written; no verbatim Cambridge text
- [ ] Authenticity ratings collected and compared
- [ ] Mapping table: every row ‘full’ after revision
- [ ] Third (pilot) passage written to the same specification

B Ethics & permissions

When: Submit in the first weeks of Sep 2027 · approval before Jan 2028

Who: You + anonymiser

Timing risk

University ethics review often takes one to three months. Approval is required **before the January pilot**. Submit as early as your supervisor allows. No data of any kind is collected before approval.

Documents to prepare

Document	Key content
Ethics application (USM human research ethics committee)	Aims, design, risks, triple-role mitigation, data handling, AI data settings
Parent information sheet + consent form	Purpose; what students do (two summaries in class); participation has no effect on grades; withdrawal until corpus-freezing date; de-identification; AI processing of typed, coded text
Student information sheet + assent form	Same, in age-appropriate language (Kurdish/Arabic versions if needed)
Principal permission letter	Permission for sessions, staff involvement and data storage
KRG / owner permission (if required)	Check before enrolment of participants
Anonymiser confidentiality agreement	Duties, storage, linkage-key destruction date
Rater 2 confidentiality agreement	Access only to coded typed scripts and 8 selected AI responses
Data management plan	Storage location (USM-approved, encrypted), access list, retention period

Consent procedure

1. The anonymiser (not you) distributes and collects forms, outside your lessons.
2. Non-participating students still write the summary as a normal class activity, but their scripts are not typed or used.
3. The anonymiser keeps the consent list; you never learn who opted in or out.
4. Students are reminded of the withdrawal deadline before typing begins.

AI data settings (confirm at the time of scoring)

Setting	ChatGPT	Claude
Chat type	Temporary chat, unpersonalised	Incognito chat
Memory / personalisation / custom instructions	Off	Off

Setting	ChatGPT	Claude
Model-training setting	Off	Off
What is entered	Coded, typed, de-identified text only	Coded, typed, de-identified text only

- [ ] Ethics approval letter received
- [ ] Principal letter signed
- [ ] KRG/owner permission resolved
- [ ] Consent and assent forms translated and approved
- [ ] Anonymiser and Rater 2 agreements signed
- [ ] Encrypted USM storage set up

C Pilot & preregistration

When: January 2028 **Who:** You, pilot colleague, anonymiser, Rater 2

Purpose: rehearse the whole chain once, on a third passage, with another teacher's students; measure timings; fix the prompt; choose the anchor scripts; then lock the plan.

1. 10–15 students in the colleague's class write a summary of the **pilot passage** under the same conditions planned for the main study.
2. The anonymiser codes and types the scripts using the transcription rules (Stage D).
3. H1 and H2 mark the typed pilot scripts independently with the study rubric.
4. Compute H1–H2 ICC(A,1) on the reading mark. **If below .60** → run a second calibration round before the main marking.
5. Score each pilot script with both tools, three passes, rubric-embedded framing (~60–90 submissions). **Time every submission.**
6. Check: the output line parses; refusals or odd answers are handled by the rules; message caps are not hit; average minutes per submission.
7. Compute the preliminary repeatability summary: mean and range of within-script absolute differences across passes, per tool.
8. Choose **three anchor scripts** (low, mid, high) from the pilot.
9. Meet the statistical adviser: expected CI width for QWK/ICC; tool × pass interaction; linear vs ordinal mixed model.
10. Fix the final prompt wording. Any change must not alter scoring instructions (supervisor + colleague decide; record in pilot log).
11. **Preregister** on OSF (contents below) before the first main writing session.

What the preregistration must contain

- [ ] Hypotheses / RQs and primary vs secondary criteria
- [ ] Sample, inclusion and exclusion rules
- [ ] Exact prompt (final wording)
- [ ] Session controls and logging
- [ ] Retry rule and valid-response definition
- [ ] Model-change decision rule
- [ ] All statistics and software (ICC forms, QWK, bootstrap, Holm)
- [ ] Mixed-model formula and fallback
- [ ] H1–H2 .60 threshold and consequences
- [ ] RQ3 selection rule incl. edge cases
- [ ] Coding scheme and evidence categories
- [ ] Contingency if discrepancies are small

- [ ] Pilot completed and logged
- [ ] Seconds/minutes per submission known
- [ ] Anchors chosen
- [ ] Prompt final
- [ ] Statistical adviser consulted
- [ ] Preregistration time-stamped on OSF

D Student writing & corpus

When: Feb – Mar 2028 **Who:** You (teacher), anonymiser

D1. Writing sessions

1. Within each class, the anonymiser randomly assigns each student to order **A→B** or **B→A** (counterbalancing). Record the seed of the random draw.
2. Session 1 and Session 2 take place on **different days**.
3. Conditions: handwritten; passage available; no dictionaries or devices; same time limit for everyone (fix the limit in the pilot; record it).
4. Scripts are collected immediately and handed directly to the anonymiser.

D2. Anonymisation and typing (anonymiser only)

Rule	How to apply it
Codes	Assign each script a random code (e.g., `K7Q2`); record passage separately. Keep the linkage list (student ↔ codes) locked.
Identifying content	Replace names, school name, teacher references, identifiable anecdotes with `[name removed]` etc.
Verbatim typing	Reproduce spelling and punctuation exactly; do not correct anything.
Crossed-out text	Keep it, marked `[crossed out]`; it is not counted or credited.
Illegible text	Mark `[illegible]`; not counted; raters note it if it affects credit.
Re-check	One week later, the anonymiser re-checks every typed script against the original.

D3. Freeze the corpus

1. After re-checking and the withdrawal deadline, the typed files are made read-only and stored in the encrypted folder.
2. The linkage list is destroyed; destruction is recorded with the date.
3. Originals stay with the anonymiser as continuing custodian.

Never

Never look at handwriting, names or the linkage list. Never type or re-check scripts yourself.

- [ ] All writing sessions done; counterbalancing record saved
- [ ] All consenting scripts typed and re-checked
- [ ] Corpus frozen; read-only copies stored
- [ ] Linkage list destroyed (date logged)

E Human marking

When: Apr – May 2028 (≥ 4 weeks after the last writing session) **Who:** You (H1) + Rater 2 (H2)

1. **Calibration:** both raters independently mark 5–8 practice scripts (previous years or public samples, not the study cohort), then meet to compare and discuss. This is not a reliability measure.
2. Generate a **separate random marking order** for each rater (record the seed).
3. Each rater marks every script with the study rubric and the reading key for that passage: **Reading /10, Writing /5**. Work alone; never discuss marks during this phase.
4. Enter marks into the marking sheet (Template T2). Double-check entry.
5. After both have finished, list scripts whose **totals differ by 3 or more**. Discuss them, but **keep the original marks** for all analyses.
6. **Lock** the human marks: save a read-only, dated copy; send a copy to your supervisor. AI scoring may not start before this.
7. Compute H1–H2 ICC(A,1) for reading, writing and total, per passage.

Preregistered rule — human benchmark

If H1–H2 ICC(A,1) falls below **.60** for a criterion on a passage, report it as a limitation and note that the benchmark for that criterion/passage did not meet the minimum. Analyses still run. If only the total falls below .60 but reading and writing meet it, only the total-score findings are qualified.

Workload	Amount
Scripts per rater	130–160
Time per script (estimate)	~5–6 minutes
Total per rater	~13–16 hours

- [ ] Calibration meeting held
- [ ] Both raters finished independently
- [ ] ≥ 3-mark disagreements discussed (originals kept)
- [ ] Marks locked and sent to supervisor
- [ ] H1–H2 ICCs computed

F AI scoring

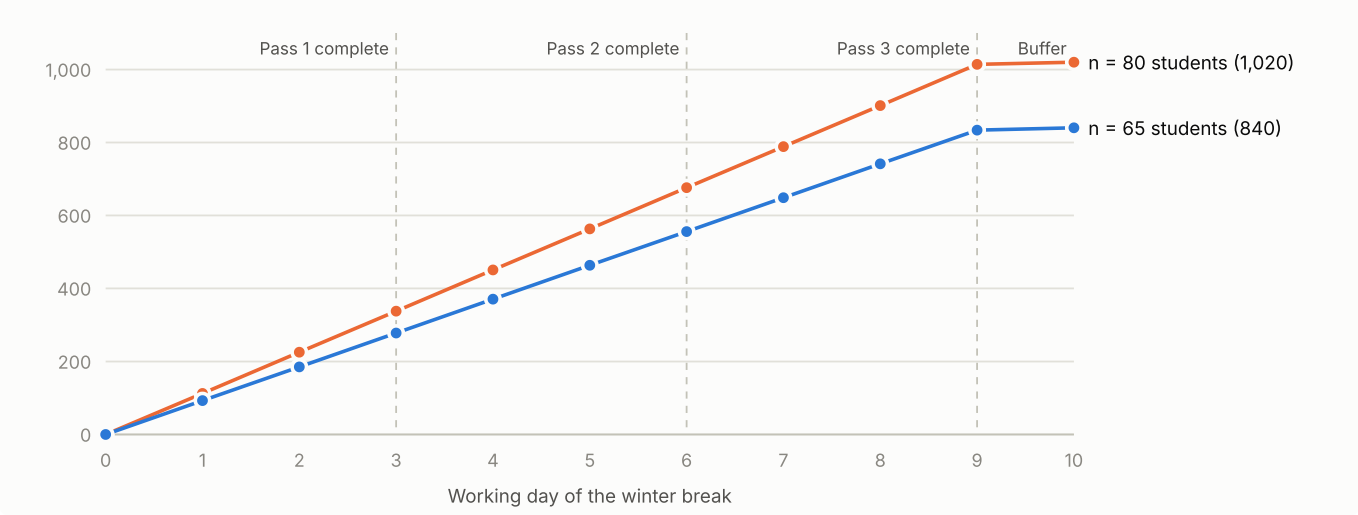
When: Winter break, Dec 2028 – Jan 2029 (10 working days) **Who:** You

F1. Before day 1

- [ ] Human marks locked (Stage E)
- [ ] Paid individual subscriptions active for ChatGPT and Claude
- [ ] Choose the latest flagship model on each paid plan; write down its exact displayed name. Select this same named model for every submission
- [ ] Settings check: memory, personalisation, custom instructions and training off; temporary (ChatGPT, unpersonalised) / incognito (Claude) chats
- [ ] Session log spreadsheet ready (Template T3); screenshot folder ready
- [ ] Randomised order lists generated for each pass (scripts and passages mixed; seed recorded)
- [ ] Prompt text final, saved as a file you copy from — never retype

F2. The ten-day plan

Days	Work	Rule check
Every day, first	Anchors: 3 anchor scripts × 2 tools = 6 submissions	Logged with anchor flag; monitored descriptively only
Days 1–3	**Pass 1**: all scripts, both tools, in blocks (e.g., ~25 ChatGPT then ~25 Claude, alternating)	Both tools scored on the same days
Days 4–6	**Pass 2**: new random order	≤ 2 passes per tool per day (here: 1)
Days 7–9	**Pass 3**: new random order	Separate sessions; fresh chat per script
Day 10	Buffer: catch-up, retries, checks	Complete all three passes inside this window



F3. What one scoring day looks like



Anchors (6) | ChatGPT block (~53) | Claude block (~53)

One scoring day at n = 80: about 112 submissions, roughly 5.5 hours at 3 minutes each (pilot confirms the real time)

F4. Exactly how to score one script

1. Open a **new** temporary/incognito chat. Confirm the selected model name.
2. Paste the prompt with the passage, task, student summary, rubric and reading key filled in (copy from your prepared files).
3. Send. Wait for the full response.
4. Copy the **full response** into the log. Parse “Reading _/10; Writing _/5” into the score columns.
5. Classify the status: valid / refusal / out-of-range / unparseable.
6. Take a screenshot; record its file name, the time, model name and reasoning mode.
7. Close the chat. Move to the next script in the list.

Preregistered rule — retries

Keep every first attempt. If the first attempt is invalid, **one** retry is allowed and logged, but retry scores enter only a separate sensitivity analysis.

Preregistered rule — model change (backup)

Selecting the same named model is the main safeguard. If the displayed model label changes between passes, analyse that tool separately by model version: RQ1 only for scripts with ≥ 2 valid scores under the same label (otherwise ‘not estimable’); RQ2 per version. A change within a pass is logged per submission and flagged. The model-change covariate is used only where it is not confounded with the pass boundary.

Never

Never edit the prompt during scoring. Never regenerate a response to ‘get a better one’. Never look up human marks while scoring. Never submit real Cambridge passages. Never continue scoring in a later holiday — all three passes must fall within this one window.

- [ ] All 3 passes × 2 tools complete
- [ ] 60 anchor submissions logged
- [ ] Every row has full response, scores, status, model name, time, screenshot
- [ ] Invalid responses and retries flagged
- [ ] Backup copy of log made the same day

G Quantitative analysis (RQ1, RQ2)

When: Jan – Aug 2029 **Who:** You (+ statistical adviser)

Reading is primary

Reading (/10) is the primary criterion; writing (/5) is secondary; the total (/15) appears only in supplementary tables. This is an analytical decision motivated by the assessment's structure.

G1. Data files

File	One row per	Columns
`human_marks.csv`	script × rater	script_id, student_id, passage, rater (H1/H2), reading, writing
`ai_scores.csv`	AI submission	submission_id, date_time, tool, model_label, reasoning_mode, pass, position, script_id, student_id, passage, anchor (TRUE/FALSE), status, retry (TRUE/FALSE), reading, writing, response_text, screenshot

`student_id` here is an anonymous study code that links a student's two scripts; it is created by the anonymiser and contains no identity.

G2. Order of analysis

#	Analysis	Primary?	Interpretation guide
1	H1–H2 agreement (ICC(A,1), QWK) per criterion and passage	Baseline	≥ .60 required; Koo & Li bands
2	RQ1: ICC(1,1) across three passes per tool × passage × criterion	Primary	<.50 poor; .50–.75 moderate; .75–.90 good; >.90 excellent
3	RQ1 supplementary: per-script SD/range; exact & adjacent agreement across passes	Supplementary	What a teacher would see on resubmission
4	RQ2: QWK and ICC(A,1), AI–H1 and AI–H2, each pass; mean and range over passes	Primary	Compare with the H1–H2 baseline; QWK ≥ .70 (Williamson et al., 2012)
5	Correction gap (AI − human mean) per pass, passage, criterion; by score level (Bland–Altman mean)	Primary	Positive = lenient; negative = harsh
6	Bootstrapped 95% CIs (2,000 resamples, student-level)	All	Interpret intervals, not point estimates
7	Mixed model: score ~ tool × passage + pass + (1|student) + (1|student:passage)	Primary	Report convergence/singularity; Holm within families. ◆ If it fails to converge: within-script SD fallback (partial, descriptive)
8	Sensitivity: mean-of-three-passes agreement; retry scores	Sensitivity	Never replaces the single-run result

G3. R code you can run

Tested logic, but always check outputs against a small hand-calculated example from the pilot before trusting them.

SET-UP

```
# install.packages(c("tidyverse","irr","lme4","Metrics"))
library(tidyverse); library(irr); library(lme4); library(Metrics)
set.seed(2029)

ai  <- read_csv("ai_scores.csv")      # one row per submission
hum <- read_csv("human_marks.csv")    # one row per script x rater

main <- ai |> filter(anchor == FALSE, status == "valid", retry == FALSE)

h <- hum |>
  pivot_wider(id_cols = c(script_id, student_id, passage),
              names_from = rater, values_from = c(reading, writing)) |>
  mutate(read_hmean = (reading_H1 + reading_H2) / 2,
         writ_hmean = (writing_H1 + writing_H2) / 2)

d <- main |> left_join(h, by = c("script_id", "student_id", "passage"))
```

RQ1 — PASS-TO-PASS CONSISTENCY

```
# RQ1: one-way random, single-measure ICC(1,1) across the three passes
icc_passes <- function(df, crit) {
  w <- df |> select(script_id, pass, all_of(crit)) |>
    pivot_wider(names_from = pass, values_from = all_of(crit)) |>
    drop_na()
  irr::icc(as.matrix(w[, -1]), model = "oneway", unit = "single")$value
}
rq1 <- main |> group_by(tool, passage) |>
  summarise(icc_read = icc_passes(pick(everything()), "reading"),
            icc_writ = icc_passes(pick(everything()), "writing"), .groups = "drop")

# Per-script SD across passes
sd_script <- main |> group_by(tool, passage, script_id) |>
  summarise(sd_read = sd(reading), sd_writ = sd(writing), .groups = "drop")
```

RQ2 — AGREEMENT AND CORRECTION GAP

```
# RQ2: single-run agreement, each pass separately, against each human rater
qwk   <- function(x, y, lo, hi) ScoreQuadraticWeightedKappa(x, y, lo, hi)
iccA1 <- function(x, y) irr::icc(cbind(x, y), model = "twoway",
                                 type = "agreement", unit = "single")$value
rq2 <- d |> group_by(tool, passage, pass) |>
  summarise(qwk_H1  = qwk(reading, reading_H1, 0, 10),
            qwk_H2  = qwk(reading, reading_H2, 0, 10),
            icc_H1  = iccA1(reading, reading_H1),
            icc_H2  = iccA1(reading, reading_H2),
            exact_H1= mean(reading == reading_H1),
            adj_H1  = mean(abs(reading - reading_H1) <= 1),
            gap_mean= mean(reading - read_hmean),
            gap_sd  = sd(reading - read_hmean), .groups = "drop")
# Report mean and range across the three passes:
rq2 |> group_by(tool, passage) |>
  summarise(across(qwk_H1:gap_sd, list(mean = mean, min = min, max = max)))
# Repeat with writing (lo = 0, hi = 5).

# Bland-Altman (per tool x passage, pass 1 shown)
ba <- d |> filter(pass == 1) |>
  mutate(avg = (reading + read_hmean) / 2, diff = reading - read_hmean)
ba |> group_by(tool, passage) |>
  summarise(bias = mean(diff), lo = bias - 1.96 * sd(diff), hi = bias + 1.96 * sd(diff))
```

STUDENT-LEVEL BOOTSTRAP

```
# Student-level bootstrap (2,000 resamples): each resampled student brings
# both scripts and all repeated scores. Script IDs are made unique per draw.
boot_ci <- function(df, stat_fun, B = 2000) {
  ids <- unique(df$student_id)
  est <- replicate(B, {
    s  <- sample(ids, replace = TRUE)
    bd <- map_dfr(seq_along(s), \(k) df |> filter(student_id == s[k]) |>
                  mutate(script_id = paste(script_id, k)))
    stat_fun(bd)
  })
  quantile(est, c(.025, .975), na.rm = TRUE)
}
# Example: CI for ChatGPT, Passage A, pass 1, QWK vs H1 (reading)
boot_ci(d |> filter(tool == "ChatGPT", passage == "A", pass == 1),
        \(x) qwk(x$reading, x$reading_H1, 0, 10))
```

MIXED MODEL

```
# Mixed model (reading; repeat for writing)
m <- lmer(reading ~ tool * passage + factor(pass) +
            (1 | student_id) + (1 | student_id:passage), data = main)
summary(m); isSingular(m)
# Options to review with the statistical adviser before preregistration:
#  - add tool:factor(pass) if pass behaviour may differ between tools
#  - ordinal::clmm() as an ordinal alternative for the bounded scales
# Holm correction within each family of contrasts: p.adjust(p, "holm")
```

H Rationale analysis (RQ3)

When: Jan – Aug 2029 **Who:** You + Rater 2 (+ supervisor check)

H1. Who does what

Step	Who	Time
Select the 8 scripts with the fixed rule (spreadsheet/R)	You	~30 min
Segment the 16 AI rationales into a numbered claim list (~6 claims each, ~100 claims)	You	~2 h
Complete the evidence record for every claim (Template T4)	You and Rater 2, separately	~8–10 h each
Apply codes (present/absent per code per claim) in Atlas.ti	You and Rater 2, separately	included above
Calculate agreement (% and Cohen's kappa per code) **before** discussing	You	~1 h
One meeting to discuss disagreements; fill the written log (Template T7). No recording.	Both	1–2 h
Supervisor reviews final coding of 3 scripts	Supervisor	~1 h

H2. Selecting the 8 scripts (4 per passage)

Slot	Rule (from scripts not yet chosen, eligible only)
1 Reading discrepancy	Largest absolute proportional reading gap (gap/10) for either tool
2 Writing discrepancy	Largest absolute proportional writing gap (gap/5) for either tool
3 Opposite direction	Largest gap on either criterion with sign opposite to the gap that selected slot 1. If that gap is 0, or tied with opposite signs, over-scoring is the reference. If none exists, take the next-largest gap and report it
4 Closest available agreement	Smallest maximum absolute proportional gap across both tools and criteria; report its actual gaps

Eligible = both tools' Pass 1 responses are valid and contain at least one claim about the summary. Ties are broken alphabetically by script code. Selection is done on the score sheet **before** reading any rationale.

```
# RQ3 selection (Pass 1, rubric-embedded), proportional signed gaps
p1 <- d |> filter(pass == 1) |>
  mutate(g_read = (reading - read_hmean) / 10,
         g_writ = (writing - writ_hmean) / 5)
# Eligible = both tools valid with a usable rationale (mark manually: usable == TRUE)
elig <- p1 |> group_by(script_id) |> filter(n() == 2, all(usable)) |> ungroup()
# For each passage: 1) max |g_read|  2) max |g_writ|  3) max |gap| opposite in sign
# to the gap that selected script 1 (zero or tie -> treat over-scoring as reference)
# 4) smallest max |gap| over both tools and both criteria. Ties: alphabetical script_id.
```

H3. Evidence judgement for every claim

Category	Use when	Example
Supported	The student text and source confirm the claim	"Mentions genetic errors" — and it does
Contradicted	The claim is demonstrably false	Credits a point that is absent → code as source-fidelity error
Insufficient explanation	Too general to show the basis of the mark	"Clear and fluent" next to a reading mark
Ambiguous	Evidence allows more than one defensible reading	Partial paraphrase of a content point

Coding rule

A short or incomplete rationale is **not** mis-scoring. A contradicted claim is an **evidence error** (e.g., source-fidelity). **Construct confusion** is coded only when one criterion is demonstrably used to justify the other criterion's mark.

H4. Codebook

Code	Definition	Example claim
Source-fidelity error	AI credits a point the student did not include, misses one they did include, or fails to detect a meaning-changing omission/distortion	"Covers the cost of cloning" when the summary never mentions cost
Construct-irrelevant rewarding	AI rewards length, sophisticated vocabulary or style not supported by the descriptor for the criterion being marked (fluency under Writing is legitimate)	"Reading 9: impressive vocabulary"
Treatment of non-standard but meaningful phrasing	AI penalises meaning-preserving non-standard wording on grounds the descriptor does not support, where a human credits it	Reading lowered for "the animals was many died" despite correct content
Rubric misapplication / construct confusion	AI explicitly uses one criterion's qualities to justify the other criterion's mark	"Reading 8/10 because the summary is fluent and well organised"
Own-words transformation	AI judges paraphrase differently from humans; ignores synonym vs structural reformulation; fails to penalise near-copying under Writing	Writing 5 for a summary with lifted sentences

Each code is a separate present/absent decision for each claim; one claim may carry more than one code. Both coders use the identical pre-segmented claim list.

How to report RQ3

These are 8 purposively selected scripts: findings are **illustrative evidence** about how problems arise, not estimates of how often they occur. If no problematic patterns appear, report it as a non-finding in the selected cases — not as proof that problems are absent.

I Synthesis, papers & thesis

When: Jan 2029 – Aug 2031 **Who:** You + supervisor

I1. Synthesis (what to write)

1. Summarise consistency and agreement by tool, passage and criterion (RQ1–RQ2).
2. Describe the rubric-related concerns illustrated by the selected cases (RQ3).
3. List the questions needing further validation before classroom adoption (e.g., teacher studies, more passages, free tiers, other tools).

Do not claim

That the workflow 'works'; when a teacher should trust a mark; which condition has the 'greatest risk' (8 cases cannot rank prevalence); that any pattern happens *because* students are multilingual.

I2. Papers

Paper	Content	Target journals	When
1	Pass-to-pass consistency and passage dependence (RQ1)	Language Testing; Language Assessment Quarterly	Draft by mid-2029
2	Human–AI agreement and the scoring correction gap, with rationale evidence (RQ2, RQ3, synthesis)	Assessing Writing; Computers and Education: AI	Draft by late 2029
3 (optional)	Reading–writing construct separation in AI rationales	Journal of Second Language Writing; TESOL Quarterly	Only if RQ3 data are deep enough

I3. Suggested thesis structure

Chapter	Content	Source
1 Introduction	Background, problem, aims, RQs, significance, scope	Proposal Ch. 1
2 Literature & framework	Summary writing, L2 writers, AI scoring evidence, validity argument, Table 2.1	Proposal Ch. 2 (updated)
3 Methodology	Everything in Stages A–H, with deviations	Proposal Ch. 3 + logs
4 Results: RQ1	Consistency	Paper 1
5 Results: RQ2	Agreement and correction gap	Paper 2
6 Results: RQ3	Rationale analysis	Paper 2/3
7 Discussion & synthesis	Validity inferences, implications, limitations, future research	New

5 Supervisor questions & answers

Questions a supervisor, panel or examiner is likely to ask, with answers consistent with the proposal. Read them aloud before meetings; adapt the wording to your own voice.

Rationale & contribution

Q. What is the original contribution of this PhD?

Not a new theory. The contribution is empirical and analytical: evidence on consumer-access AI scoring of a source-based adolescent task with **separate reading and writing marks**, plus a reusable **claim-level evidence-checking procedure** showing when score agreement gives misleading reassurance. To my knowledge no published study combines these elements (Table 2.1).

Q. Sawaki et al. (2025) already did repeated runs and rationale analysis. How are you different?

Sawaki et al. used checklist items on topic sentences with Japanese undergraduates. I use separate holistic reading and writing marks on an IGCSE task, adolescent writers, consumer-subscription access, two passages, and two independent coders checking each AI claim against the source, the student text and the rubric.

Q. Why should anyone care if the specific models will be outdated?

The measurement problem outlives the models: whether agreement and defensible justification are separate questions, and how to detect criterion-level problems. The documented corpus can also be re-scored by future models.

Q. Isn't the 'scoring correction gap' just the AI–human difference?

Yes — numerically it is an established comparison. It is an organising label; any originality lies in the criterion-level, evidence-checked analysis attached to it, not in the number.

Design

Q. Why consumer subscriptions and not the API?

Because that is how teachers actually use these tools. It is a deliberate methodological choice motivated by the use context. API studies answer what models can do under control; this study answers what teachers get.

Q. Why not the free versions?

The study is scoped to paid individual plans. Free tiers can differ in model and usage limits, so findings are not assumed to apply to them. That is a scope decision stated in Section 1.6.

Q. Why three passes and not two or five?

Three gives a within-script variance estimate from three data points instead of one difference, while staying feasible (~42–51 hours of scoring). More passes would improve precision but exceed the time available.

Q. Why only one prompt framing?

To keep the design focused and the workload realistic. Prompt sensitivity is acknowledged as a limitation and left for future research.

Q. Why two passages?

To see whether scoring properties hold across texts within the same task type. Two passages allow comparison between those two; they do not establish generalisability to all IGCSE texts.

Q. Why write your own passages instead of using past papers?

No copyright issue, unlikely to be in training data, no published mark scheme to memorise, and full control of content points. Authenticity is checked against real passages by independent colleagues.

Q. Your rubric is not Cambridge's. Does that weaken the study?

It means the study evaluates AI scoring against an IGCSE-aligned rubric. The criterion-mapping table (Appendix C), independently reviewed by two colleagues, shows each descriptor preserves the Cambridge criterion without moving qualities between reading and writing.

Q. What if the model changes during scoring?

I select the same named model for every submission. If the label still changes, the preregistered rule analyses by model version and reports within-version repeatability as not estimable where fewer than two passes share a label. An unchanged label cannot guarantee an unchanged service; this is acknowledged.

Q. Why score everything in one winter break?

All three passes must fall in one bounded window so short-term repeatability is not mixed with longer-term service changes. Ten working days at about 4.5–5.5 hours a day is feasible in a two-week break; the pilot confirms the timing.

Sample & statistics

Q. Is 65–80 students enough?

For estimating agreement with useful precision, yes, but intervals will be moderately wide. The 780–960 AI scores are repeated measures, not independent observations; precision depends mainly on students. Results are interpreted through bootstrapped intervals, and expected width is discussed with the statistical adviser before preregistration.

Q. Why bootstrap by student?

Each student contributes two scripts and many repeated scores. Resampling students keeps that dependence intact, so intervals are not falsely narrow.

Q. Which ICC forms and why?

ICC(1,1) one-way for pass-to-pass consistency (passes are interchangeable occasions); ICC(A,1) two-way absolute agreement for AI vs each human rater, so level differences count as disagreement.

Q. Why QWK as well as ICC?

QWK is the standard metric in automated scoring and allows comparison with the Williamson et al. (2012) .70 benchmark; ICC adds an absolute-agreement view. Both are reported against the human–human baseline.

Q. Why is reading primary?

It carries two-thirds of the marks and has the longer scale, which may allow more precise estimates. It is an analytical decision, not a requirement of the mark scheme. Writing is reported fully as secondary.

Q. What does the mixed model add?

Systematic tool, passage and pass effects with students as random effects. The adviser reviews whether a tool × pass interaction or an ordinal model is needed. A preregistered descriptive fallback exists if it does not converge.

Q. What if the two human raters disagree a lot?

Calibration first; a second calibration if pilot reading ICC < .60. If main-study ICC is still below .60 for a criterion/passage, that benchmark is reported as weak and the related findings are qualified.

RQ3 & qualitative

Q. Are 8 scripts enough?

For an illustrative component, yes. They are chosen by a fixed rule to cover both passages, both criteria, both directions of disagreement and a close-agreement case. Findings are not prevalence estimates.

Q. Why include a close-agreement case?

Agreement can rest on reasoning the rubric does not support. This case tests whether the justification is sound even when the mark matches.

Q. Do AI rationales show how the model actually scored?

No. They are reported justifications, not verified accounts of the internal process. They are analysed as what a teacher would read and act on.

Q. How do you avoid calling a brief rationale an error?

Four evidence categories: supported, contradicted, insufficient explanation, ambiguous. Missing explanation is not mis-scoring; construct confusion requires explicit cross-criterion misuse.

Q. How reliable is your coding?

Two independent coders, identical pre-segmented claim list, present/absent per code per claim, percentage agreement and Cohen's kappa calculated before discussion, a written disagreement log, and a supervisor check of three scripts.

Q. Can you say the AI disadvantages multilingual learners?

No. There is no comparison group. I can describe how the tools treat non-standard but meaningful phrasing within this multilingual sample, not attribute patterns to language background.

Ethics & feasibility

Q. You are teacher, researcher and rater. How is bias controlled?

External anonymiser handles consent and typing; I never see names or handwriting; identifying content removed; random marking order; marks locked before AI scoring; four-week delay; independent second rater. Familiarity cannot be fully removed; this is acknowledged.

Q. Could students feel pressured to participate?

Consent is collected by the anonymiser, not by me; participation has no effect on grades; I never know who opted out because non-participants write as a normal class activity.

Q. Is it ethical to put student writing into ChatGPT and Claude?

Only coded, typed, de-identified text is entered, with training off and temporary/incognito chats. Provider terms are re-checked at the time of scoring and described in the ethics application.

Q. Can you really do 42–51 hours of scoring in two weeks?

Yes, at about 4.5–5.5 hours a day with a buffer day; the pilot measures real time per submission. It is planned for the winter break specifically because summers are for supervision visits.

Q. What happens if the ethics approval is late?

The pilot and writing shift by a month or two; Feb–May has some slack. The application is submitted in the first weeks of candidature to minimise this.

Q. Who else is needed, and for how long?

Rater 2 (~25–30 h), an anonymiser (~15–20 h), a pilot colleague (one lesson) and 2–3 reviewers (~2 h each). The team is intentionally small.

Outcomes

Q. What if AI agrees closely with the humans?

That is reported as a finding: under these conditions, consumer AI scoring of this task agreed closely with human judgement. The contribution then lies in establishing the agreement level under ecological conditions.

Q. What will teachers actually get from this?

Descriptive evidence on consistency, agreement and justification quality for this task, informing further evaluation of AI as a second marker. It does not specify when a teacher should trust a mark; that needs studies with teachers.

Q. What are your planned publications?

Two primary papers (RQ1; RQ2 + RQ3) in assessment/EdTech journals, and an optional third on construct separation if RQ3 data are deep enough.

6 All decision rules in one place

Situation	Preregistered decision
Pilot H1–H2 reading ICC(A,1) < .60	Second calibration round before main marking
Main H1–H2 ICC(A,1) < .60 (criterion/passage)	Report as limitation; qualify those findings; analyses still run
Human totals differ by ≥ 3	Discuss after independent marking; original marks always used
First AI attempt invalid	One logged retry; retry scores only in sensitivity analysis
Missing human mark or valid AI score	Complete-case by script for that condition; report attrition
Displayed model label changes between passes	Analyse by version; RQ1 needs ≥ 2 valid scores per label, else 'not estimable'; covariate only if not confounded with pass
Label changes within a pass	Log at submission level; flag; inclusion decided at preregistration
Anchor scores drift	Descriptive only; never exclude data on anchors
Prompt/parsing change after pilot	Only if it does not alter scoring instructions; supervisor + colleague decide; record
Mixed model fails to converge	Within-script SD fallback, reported as partial
RQ3 slot 3 reference gap is 0 or tied	Over-scoring is the reference direction
RQ3 script ineligible	Skip to next; report number skipped
RQ3 tie	Alphabetical by script code
Coders disagree	Kappa first; then discuss; written log; unresolved cases reported
No problematic RQ3 patterns	Report as non-finding in selected cases

7 Risk register

Risk	Likelihood	Impact	Prevention / response
Ethics approval late	Medium	High	Submit in first weeks; drafts ready before enrolment; shift pilot/writing by 1–2 months if needed
Rater 2 unavailable	Low	High	Agree dates early; identify a reserve IGCSE colleague
Low consent rate	Medium	Medium	Clear information sheets; anonymiser-led consent; report achieved n
Weak human agreement	Medium	High	Calibration; second round if pilot < .60; report and qualify
Message caps slow scoring	Medium	Medium	Pilot tests caps; alternate tools in blocks; buffer day
Model retired mid-scoring	Low	Medium	Backup model-change rule; log everything
Scoring not finished in winter break	Low	High	Pilot timing; ~5 h/day plan; buffer day; protect the break from other duties
Mixed model does not converge	Medium	Low	Adviser review; ordinal alternative; descriptive fallback
Too few eligible RQ3 scripts	Low	Medium	Eligibility rule; report shortfall
Data loss	Low	High	Daily backup to encrypted USM storage + one offline encrypted copy
Identification of a student	Low	High	Anonymiser removes identifiers; you never see originals

8 Templates & forms

T1. AI PROMPT (DRAFT — FINALISE AFTER THE PILOT)

```
Below are a reading passage, a summary task, and a student's summary written in response to the
task.

PASSAGE: [PASSAGE]
TASK: [TASK]
STUDENT'S SUMMARY: [SUMMARY]

The student's summary may contain text marked [crossed out] or [illegible]. Text marked [crossed
out] is excluded from the word count and is not credited. Text marked [illegible] is likewise
excluded from the word count; note any [illegible] passages when they affect the creditability of
content.

Mark this summary. It is marked out of 10 for reading and 5 for writing.
Use the following marking rubric: [RUBRIC]
Award the reading mark holistically using the level descriptors in the rubric above. The reading key
below lists indicative content points to help you understand what the passage offers; treat it as a
guide, not a checklist to convert into marks. Other valid ideas not listed here may be credited.
[READING KEY]

Give your marks on the first line in exactly this form: Reading _/10; Writing _/5
Then give a short reason for your marks, with reference to the level descriptors.
```

T2. HUMAN MARKING SHEET

script_id	passage	rater	reading (0–10)	writing (0–5)	illegible note	date
K7Q2	A	H1	7	3	—	2028-04-12
…						

T3. AI SESSION LOG (ONE ROW PER SUBMISSION)

Field	Example
submission_id / date_time	S0457 / 2028-12-27 10:41 (UTC+3)
tool / model_label / reasoning_mode	ChatGPT / [displayed name] / [as displayed]
pass / position / anchor	2 / 87 / FALSE
script_id / passage	K7Q2 / A
settings check	yes
full response	(paste whole text)
reading / writing	7 / 3
status / retry	valid / FALSE
screenshot / anomalies	S0457.png / none

T4. EVIDENCE RECORD (ONE ROW PER CLAIM)

Claim #	Script / tool	AI claim (verbatim)	Criterion	Student wording + source evidence	Applicable descriptor	Judgement	Codes (present)
014	K7Q2 / Claude	“covers rising costs”	Reading	No mention of cost; source para 3	Reading L4	Contradicted	Source-fidelity

T5. PASSAGE-AUTHENTICITY RATING FORM (1 = MUCH EASIER/LESS ... 5 = MUCH HARDER/MORE)

Passage (unlabelled)	Difficulty	Text-type fit	Content-point density	Vocabulary level	Length appropriateness	Comment
P1						
P2						
...						

T6. CRITERION-MAPPING REVIEW FORM

Row	Cambridge criterion	Study descriptor	Full / partial / absent	Comment
R1	Range of relevant ideas understood			
...				

T7. CODING DISAGREEMENT LOG (WRITTEN; NO RECORDING)

Claim #	Coder 1	Coder 2	Final decision	One-line reason
014	Contradicted	Ambiguous	Contradicted	Summary contains no reference to cost
022	Supported	Contradicted	Unresolved	Descriptor allows both readings

9 Data management & file naming

FOLDER STRUCTURE (ENCRYPTED USM STORAGE)

```
PhD_AIScoring/
  00_admin/          ethics letters, permissions, agreements, deviation notes
  01_instruments/    passages, reading keys, rubric, prompt (versioned)
  02_pilot/          pilot scripts, marks, AI log, timing, pilot report
  03_corpus/         frozen typed scripts (read-only)       [no identities]
  04_human_marks/    calibration, H1, H2, locked file
  05_ai_scoring/     session log, screenshots/, order lists, seeds
  06_analysis/       R scripts, outputs, figures
  07_rq3/            selection sheet, claim list, evidence records, Atlas.ti project, log
  08_writing/        papers, thesis chapters
```

NAMING RULES

Item	Pattern	Example
Typed script	script_id_passage.txt	K7Q2_A.txt
AI screenshot	submission_id.png	S0457.png
Versioned document	name_vX_YYYYMMDD	rubric_v3_20271120.docx
Locked file	name_LOCKED_YYYYMMDD	human_marks_LOCKED_20280530.csv

Back up every day you work

One copy on encrypted USM storage, one offline encrypted copy. Never store identities anywhere in this structure.

FINAL CHECKLIST BEFORE SUBMITTING THE THESIS

- [ ] Every deviation from the preregistration is listed with date and reason
- [ ] All locked files unchanged (compare dates/hashes)
- [ ] Corpus documentation ready for future re-scoring (subject to consent and policy)
- [ ] Retention period follows the USM research data policy
- [ ] AI use in writing disclosed according to USM and journal rules

End of handbook · Keep this document with the proposal and update the version number whenever a rule changes.